



Supplementary Materials for

Correcting the logic behind the 2026 US vehicle emissions standard rollback

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Materials and Methods
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Other Supplementary Material for this manuscript includes the following:

Data S1

Materials and Methods

A. Numerical Estimates

Guided by our framework, which we will develop below, we propose various adjustments to the cost-benefit analysis in DRIA Tables 2 and 3. First, we propose a different numerical split between categories (i), (ii) and (iii) than either of the previous two EPAs. Second, we suggest a different fuel price forecast. Third, we propose a numerical adjustment to the technology cost curve to reflect the possibility for attribute changes. We conclude with an overview of how these three changes affect the net benefits calculation for the current proposal.

A.1 Fuel-Savings Valuation

Category (i)

How much are consumers willing to pay for a car that saves \$1 in fuel? We will assume this number accounts for both the fuel and any (visible or invisible) technologies that the consumer believes will come packaged with this new car. EPA suggests consumers would pay for the portion of fuel saving that arrives in the first 2.5 years. This works out to around \$0.21-\$0.25 of the \$1 of fuel savings depending on vehicle life and discounting assumptions. We note that this number is based on limited evidence, but in the absence of publicly available studies of this value, we illustrate the framework using this number.

The 2.5 year assumption has been used in multiple vehicle GHG and fuel economy regulations, and repeated in several public comments. Those that cite evidence for this assumption can be traced back to three sources. The first is a 2015 National Academies study that states that “[d]uring its information-gathering process, the committee found that auto manufacturers perceive that typical consumers would pay upfront for only one to four years of fuel savings.” (NASEM 2015, p. 315). The second is a paper by David Greene (who served as a committee member on the National Academies study) that states that “[t]he author has discussed fuel economy issues with knowledgeable staff of most of the world’s major automobile manufacturers over the past two decades. Manufacturers consistently report that consumers require a simple payback period of 2-4 years for improvements in fuel economy. These perceptions are based on internal research and market analyses that are not made public for competitive reasons.” (Greene 2011, p. 611). The third is a survey conducted in 2002 by the Opinion Research Corporation for the National Renewable Energy Laboratory that asked respondents “How soon, in years, would the fuel savings have to pay back the additional cost to persuade you to buy the higher fuel-

economy option?” (Kubik 2005, p. 12). The mean response was 2.9 years, with a standard deviation of approximately 2.2 years.

Provided that the evidence for the 2.5 year assumption is based on older studies that are largely proprietary manufacturer studies with unknown methods, we recommend that additional studies estimate this payback period for contemporary consumers and economic conditions.

How do we arrive at \$0.21-0.25? We implement a simple life cycle simulation to quantify how much of a vehicle’s lifetime fuel savings occur in the first 2.5 years. Assumptions mirror EPA/OMEGA conventions where possible: a 15-year service life (close to OMEGA’s survival midpoint), first-year VMT of 16,000 miles with an annual VMT decay of 3.9%, baseline fuel economy of 27.1 MPG, and gasoline at \$3 per gallon. We then raise fuel economy to 30 MPG, compute annual fuel spending with and without the improvement, and form the ratio of savings accrued in the first 2.5 years to total lifetime savings under several discounting conventions.

Because incremental fuel savings are proportional to miles driven, the share in the first 2.5 years is independent of the MPG change; only the mileage profile, lifespan, gas prices, and discounting matter. With front-loaded driving (higher miles driven in year one and 3.9% annual decline), the initial years account for more than a simple 2.5/15 share. The resulting shares are:

No discounting of numerator or denominator: 0.210 ($\approx 21.0\%$)

Discount both numerator and denominator: 0.245 at 3% and 0.293 at 7%.

Do not discount numerator, discount denominator: 0.251 at 3% and 0.308 at 7%.

Category (ii)

Missing estimate of pure undervaluation/internalities such as inattention. Assigning all undervaluation to missing costs, and none to inattention (a term that is used to refer to undervaluation of fuel savings due to misperceptions about future fuel use or price, present bias, or other cognitive biases), is not supported by the economics literature. Table 1 of the DRIA includes several important studies of consumer inattention to fuel economy (U.S. EPA 2025). The DRIA first mentions that estimates from older studies span a wide range and are therefore inconclusive. Later, the report argues that the range is narrower, using more recent studies with more modern econometric methods, pointing to consumers valuing a large proportion of fuel savings and perhaps even all—hence leaving little scope for inattention. However, DRIA Table 1 shows that there is a wide range of values in the literature; there is not a consensus on full attention. The table includes important studies but leaves out other, and more recent, equally

credible estimates, which predominantly find substantial undervaluation of fuel savings due in whole or part to inattention.

We propose adding the following studies and base estimates of consumer valuation of fuel economy; the corresponding estimate of inattention is one minus the percentages listed below of consumer valuation of fuel economy:

Gillingham, Houde, and van Benthem (2021): 16-39% at a 4% discount rate (13-33% at 1%; 18-46% at 7%; 23-58% at 12%).

Grigolon, Reynaert and Verboven (2018): 91% at a 6% discount rate. (Note: this is estimated in the European context and the U.S. EPA has not typically included estimates from outside the United States.)

Leard, Linn, and Springel (2025): 9-56% across income groups, with a mean of 22% at 3-5% discount rates.

Leard, Linn, and Zhou (2023) is in the RIA but the authors argue they should be cited as “54% at a 1.3% discount rate” and do not agree with EPA’s addition of the 73%, which refers to a calculation in the appendix that uses outdated data.

The view on consumer inattention and undervaluation amongst economists has thus been changing, and the EPA should consider the full body of evidence. Table S2 shows the best-available estimates in the literature at various discount rate assumptions. These studies either study “otherwise identical vehicles” or include detailed controls for vehicle characteristics, so the

reported valuation ratios (e.g., 58% for Allcott and Wozny (2014)) imply that the balance is attributable to inattention (category (ii); e.g., 42% in for Allcott and Wozny (2014)).

Table S1: Estimates of consumer attention to fuel savings.

Study	Future fuel price assumption	Vehicle types	Discount rate assumption			Gap provides estimate of:
			1-3%	5%	6-7%	
Busse et al. (2013)	Gasoline price at time of sale	New and used cars and light trucks	71	78	81	(ii)
Allcott and Wozny (2014)	Average of gasoline price at time of sale and oil futures-based forecast	Used cars and light trucks	58	62	66	(ii)
Sallee et al. (2016)	Gasoline price at time of sale	Used cars and light trucks	76	101	113	(ii)
Grigolon et al. (2018)	Gasoline price at time of sale	New cars and light trucks	71	86	91	mostly (ii)
Gillingham et al. (2021)	Gasoline price at time of sale	New cars and light trucks	23	30	32	(ii)
Leard et al. (2023)	Gasoline price at time of sale	New cars and light trucks	54	69		mostly (ii)
Leard et al. (2025)	Gasoline price at time of sale	New cars and light trucks	22	27	30	mostly (ii)
Average			54	65		

Notes: Relative to DRIA Table 1 (U.S. EPA, 2025), we only keep one row of estimates per table, which represent the preferred specification presented by the authors of those papers. Allcott and Wozny is the average of the two rows in DRIA Table 1. Leard et al. (2023)'s 54-73% range from DRIA Table 1 in the 1-3% discount rate category is included here as 54%; see their public comment nr. EPA-HQ-OAR-2025- 0194-3006. Some values are interpolated to match the discount rate columns chosen by the EPA. The 10% discount rate column is not included as that

falls far outside the range of the literature. Table shows estimates of consumer attention; the corresponding estimate of inattention is one minus these percentages.

Discount rate. DRIA Table 1 seems to argue a central discount rate of approximately 6% (the average between the 5% and 6-7% columns). We propose that the fuel economy valuation ratio be estimated using a discount rate of 3-4% (the average between the first and the second column of Table S2). The studies listed in the table have assumed discount rates in the 1.3-6% range. Using a relatively low discount rate is justified, especially in studies studying new-vehicle purchases, as new-vehicle buyers have access to car loans, and can likely borrow at low rates in general. In the period of the first published studies from Table 1, for example, auto loan rates were around 2.5% in real terms. More recently, auto loans have had similar real rates (6-7% nominal with 3-4% inflation) (Statista 2025). Taken together, a discount rate range of 3-4% appears most in line with the range of studies and interest rates during their sample periods.

Exact vs. approximate valuation ratios. Gillingham et al. (2021) discuss that the details of how the valuation ratio is calculated really matter. We refer to their Sections IV.C and IV.D for details, but note that the root of the issue lies in the issue that the average of a set of (fuel-economy valuation) ratios for individual vehicle models can differ starkly from the ratio of the average change in vehicle transaction prices and the average change in fuel economy for all vehicle models. Gillingham et al. (2021)'s analysis suggests that the estimates from studies in the bottom half of their Table 8 are inflated by a factor of ~ 2 if the bias is constant across data

samples/periods. This is a caveat for some often-cited estimates pointing to full valuation. Table S3 adjusts the valuation ratios of the relevant studies downwards.

Table S2: Estimates of consumer inattention to fuel savings, with approximate valuation ratios adjusted downwards.

Study	Future fuel price assumption	Vehicle types	Discount rate assumption			Gap provides estimate of:
			1-3%	5%	6-7%	
Busse et al. (2013)	Gasoline price at time of sale	New and used cars and light trucks	35	39	41	(ii)
Allcott and Wozny (2014)	Average of gasoline price at time of sale and oil futures-based forecast	Used cars and light trucks	58	62	66	(ii)
Sallee et al. (2016)	Gasoline price at time of sale	Used cars and light trucks	76	101	113	(ii)
Grigolon et al. (2018)	Gasoline price at time of sale	New cars and light trucks	36	43	46	mostly (ii)
Gillingham et al. (2021)	Gasoline price at time of sale	New cars and light trucks	23	30	32	(ii)
Leard et al. (2023)	Gasoline price at time of sale	New cars and light trucks	27	35		mostly (ii)
Leard et al. (2025)	Gasoline price at time of sale	New cars and light trucks	11	14	15	mostly (ii)
Average			38	46		

Conclusion on the inattention estimate. Taken together, considering the range of high-quality studies of fuel economy valuation, and applying reasonable discount rates in 3-4% range, the valuation ratio averages at around 54-65%. When making the adjustment for exact vs.

approximate valuation parameters, the average valuation ratio equals 38-46%. We thus propose that the typical finding in the context of conventional vehicles is that consumers value future fuel savings at about 42 cents to the dollar, and thus enjoy 58 cents in ex-post savings. This contrasts starkly with EPA's 2025 DRIA assumption of zero, and also contrasts with EPA assumptions from prior rulemakings.

Category (iii)

To provide an estimate of missing costs, we need to consider how consumers value the change in attributes that come along with a more fuel-efficient car—gasoline, hybrid, or electric.

The literature has shown that there is significant heterogeneity of consumer preferences for both observed vehicle attributes (e.g., acceleration) and unobserved vehicle attributes (e.g., other aspects of a vehicle that change in cases where stringent regulation leads to converting it to a BEV) that affect consumer valuation (Helveston et al., 2015; Forsythe et al., 2023). Some consumers have a large negative valuation of BEVs compared to conventional gasoline vehicles, while others have a large positive valuation of BEVs relative to conventional vehicles. At the margin, there are consumers willing to switch to a BEV or trade off other attributes of their vehicle if given only a small nudge. These closer-to-indifferent consumers will switch to BEVs under a regulation. Thus, the right metric to consider is not the average consumer valuation of BEVs, but the change in valuation of the switching consumers for the vehicle they would have purchased with and without the regulation.

Forsythe et al. (2023) estimated consumer valuation of model year 2022 BEVs where the manufacturer also produced a counterpart conventional gasoline version of the vehicle. Using these estimates, we calculate the missing costs (category iii) for BEV switchers to reach the increases in BEV share that EPA projected for that vehicle's segment (i.e., sedan/wagons or CUVs/SUVs) in the model year model year 2027-2032 GHG standards.

Using a 3% discount rate, missing costs were 22-27% of the fuel savings for the SUV models and 17-31% of the fuel savings for the passenger car models. The variation of these estimates is influenced by the BEV range. BEVs at the upper end of the percentage estimates had a shorter range (i.e., 110 miles for passenger cars and 220 miles for SUVs), and those at the lower end had a longer range (i.e., 280 miles for passenger cars and 240 miles for SUVs). Using a 7% discount rate, missing costs were 27-33% of the fuel savings for the SUV models and 21-38% of the fuel savings for the passenger car models. This contrasts with the assumption in the 2025 DRIA that missing costs are 79%.

The estimated missing costs above may be upper bounds for three reasons: (1) Technology advances over time are expected to improve BEV attributes from model year 2022 and therefore reduce missing costs for model year 2027-2032 vehicles. (2) Consumers are not restricted to only consider the BEV version of the vehicle they would have purchased without the regulation. They have many different vehicles they can choose from, and they will only switch to those vehicles if they value them more. (3) As the share of BEVs increase under a regulation, the variety of BEVs

produced increases, allowing consumers more options to choose a vehicle that, for them, has smaller attribute tradeoffs.

Taken together, our suggested ranges for categories (i), (ii), and (iii) are 21-25%, 54-62%, and 17-31%, respectively. While determined separately, the midpoints of these ranges sum up to 100% almost exactly. We propose the EPA use \$0.22, \$0.55, and \$0.23 for consumer valuation, inattention, and missing costs, respectively.

A.2 Fuel Price Forecasts

The proposed rule's RIA includes various assumptions about future fuel costs, which are a crucial parameter for calculating the benefits of improved vehicle efficiency. Tables 2 and 3 in the DRIA start with the fuel cost assumptions used in the EPA's analysis of the 2024 multi-pollutant rule, which use the Annual Energy Outlook (AEO) 2023 gasoline price projections by the U.S. Energy Information Administration (2023). Next, the DRIA discusses how the AEO 2025 has a lower reference case projection of fuel prices (DRIA Figure 1). As explained in the AEO 2025, this lower gasoline cost projection is due to the expected increase in vehicle efficiency due to the 2024 multi-pollutant rule, as well as several other regulations that all promote electric vehicles relative to gasoline vehicles. However, the current policy rollbacks will increase demand for gasoline and thus remove the rationale for the assumed decrease in gasoline prices. AEO 2025 also has an "Alternative Transportation" case where these regulations are removed. This is the relevant case for the proposed rule and we suggest it be the reference case, perhaps along with sensitivity analysis based on the High Oil Price and Low Oil Price scenarios.

The AEO 2025 "Alternative Transportation" projection shows gasoline prices very similar to, but just below, those in the AEO 2023 projection—by approximately \$0.16 per gallon on average (\$0.22 per gallon less in 2027 and \$0.14 less in 2038)--which justifies slightly lower fuel-savings benefits from regulation than in the 2024 multi-pollutant rule analysis. However, the 2025 DRIA relies heavily on a "low liquid fuel prices" scenario (column 3 in Tables 2 and 3). This scenario subtracts \$0.25 per gallon for diesel and \$1.00 per gallon for gasoline, without any justification for these magnitudes. This leads to a substantial underestimate of the increase in fuel costs associated with the proposed rule.

A.3 Technology Costs

As noted, EPA's 2024 analysis of greenhouse gas standards assumed vehicle quality and attributes would remain fixed. The technology cost curves were designed to reflect the cost of improving fuel efficiency while holding other performance attributes like horsepower and acceleration constant. Consistent with this assumption, the analysis did not assign any "missing costs" to consumers from potential worsening of vehicle attributes (e.g., reduced driving range) or "missing benefits" from potential benefits (e.g., improved low-speed torque). One may reasonably question whether compliance would in fact have been achievable without changing

attributes in ways that are viewed as costly by consumers, but the 2024 EPA analysis was internally consistent in the idea that vehicle driving experience and utility would not be compromised.

In contrast, the 2025 EPA analysis assigns large missing costs to reflect consumers' distaste for vehicle attribute changes. However, the supporting analysis still uses the old cost curves that reflect the costs of improving fuel economy while preserving driving experience and utility of vehicles. This inconsistent combination makes the cost of regulation appear high (by requiring technologies that do not allow attribute tradeoffs), and it makes the benefits of regulation appear small (by assuming that most fuel savings get negated by missing costs associated with worsened vehicle attributes).

If EPA wishes to model missing costs of regulation associated with worsened vehicle attributes, then we suggest it makes two changes. The first is to model technologies that include tradeoffs between fuel savings and attributes, which would have improved both the 2024 and 2025 EPA analyses. It can also then consider technologies that save fuel without compromising other features of the vehicle. Manufacturers would then choose the profit-maximizing bundle of these technologies.

Existing estimates show that using attribute tradeoffs for conventional gasoline vehicles reduced total (producer and consumer) compliance costs of the vehicle GHG standards by 42% (Whitefoot et al. 2017). Automakers lower compliance costs by producing a mix of vehicles with different strategies when consumer preferences are heterogeneous. For example, by producing some vehicles that use attribute tradeoffs for consumers that do not mind the tradeoffs as much, and producing other vehicles that do not have tradeoffs for consumers that highly value the attributes. Future work in this area should update this analysis using the current mix of technologies and available strategies to comply with the standards.

A.4 Implications for Cost-Benefit Analysis

Updated fuel cost forecasts. The gasoline price forecast in the 2025 DRIA is somewhat arbitrarily set to the 2023 AEO forecast minus \$1.00/gallon. As explained above, we argue that the EPA should use the 2025 AEO "Alternative Transportation" forecast, which is on average \$0.16 lower than the 2023 AEO forecast. This results in a much more modest decrease in the costs of rescinding the standard between Scenarios 2 and 3 in DRIA Tables 2 and 3 equal to 16% * $(\$1,390 - \$870) = \$83$ billion (3 percent) or \$43 billion (7 percent).

Updated fuel-savings split. The DRIA shows the net present value of the monetized savings, costs, and net savings from rescinding the GHG standards at a 3 percent discount rate (Table 3) and a 7 percent discount rate (Table 2), in billions of \$2022. Scenarios 2 and 4 in these tables compare the net benefits of rescinding the standards using the 2024 EPA assumption of full fuel-savings valuation (full inattention; no missing costs) and the 2025 EPA assumption of 21% fuel-savings valuation (no inattention; \$0.79 is all in missing cost category (iii)). Both EPA

methodologies are incorrect as extreme cases; our proposed split reflects the body of evidence in the literature.

Updated technology costs. Updated estimates of the compliance cost savings of using attribute tradeoffs are needed to reflect the tradeoffs for the current mix of vehicle technologies, including PHEVs and BEVs as well as conventional gasoline vehicles. With that caveat, we show reductions in compliance costs using the existing 42% estimate in the literature.

Adjustments to cost-benefit analysis. We now correct the cost-benefit analysis using our proposed adjustments to the fuel cost scenario, the division of fuel-savings, and the technology costs. Table S3 shows the results using 3 percent and 7 percent discount rates, but as argued above, we consider the 7 percent unreasonably high. Note that Table S3 shows the costs and benefits of rescinding the standards, i.e., savings are foregone technology costs and costs are increases in fuel expenditures that are eventually valued by a—potentially initially inattentive—vehicle owner.

Starting from the EPA’s main scenario (reproduced below in Table S3, Column 1), we first estimate that applying fuel costs from the 2025 Annual Energy Outlook (AEO) leads to a \$92 billion increase in the costs of rescinding the standards (using 3 percent; and \$50 billion using 7 percent) in Column 2. We do this by taking the difference between Columns 4 and 5 in Tables 2 and 3 in the DRIA, and multiplying the difference by 0.84 (the difference between the EPA’s \$1 fuel cost reduction estimate and the AEO 2025’s \$0.16).

Next we apply our proposed division of fuel-savings: \$0.22 (direct consumer valuation), \$0.56 (inattention), and \$0.23 (missing costs). This division can be summarized in the following figure:

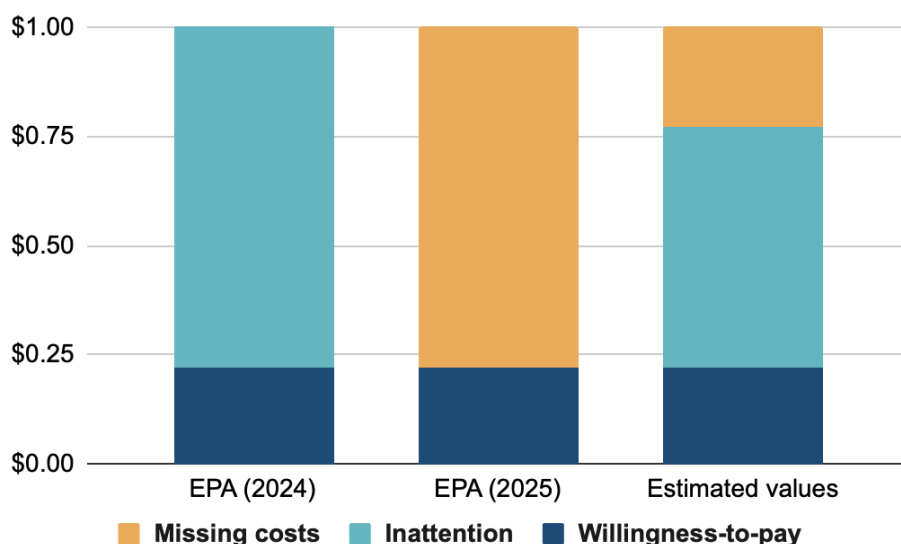


Fig. S1: Split of fuel savings into consumer willingness-to-pay, inattention, and missing costs from attribute tradeoffs. The first bar indicates assumptions in EPA’s 2024 analysis.

The second bar indicates assumptions in EPA’s 2025 analysis. The third bar shows the best available estimates we construct from literature.

The costs increase further by \$449 billion and \$227 billion, respectively (Column 3). We do this as follows:

We start with DRIA Tables 2 and 3, Columns 2 and 4. Column 2 has the 2024 EPA’s values, without the IRA and ACT. Column 4 keeps the same 2024 EPA values (including the old fuel cost forecast), but has the 2025 EPA’s fuel savings split.

Applying our proposed fuel savings split, we close the difference between the costs of rescinding the standard (Column 2 minus Column 4) by $(\$0.55 / (\$0.55 + \$0.23))$. At a 3 percent discount rate, the adjusted costs become $\$660 + (\$0.55 / (\$0.55 + \$0.23)) * (\$1,390 - \$660) = \$1,175$ billion. At a 7 percent discount rate, this number is \$581 billion.

Now, we need to move to the slightly lower fuel cost scenario from AEO (2025). We therefore subtract the difference between Columns 2 and 3 in DRIA Tables 2 and 3, and multiply this by 0.16 (the difference between the EPA’s \$1 fuel cost reduction estimate and the AEO 2025’s \$0.16). This reduces the cost of rescinding the standards to \$1,092 billion and \$538 billion, respectively, the values shown in Column 3 of Table S3 below.

Finally, in Column 4 of Table S3, we multiply the technology cost savings from rescinding the standards by $(1 - 0.42)$, to correct for the logical inconsistency of measuring technology costs

while holding vehicle performance constant. This reduces the technology cost savings from rescinding the standards by \$433 and \$269 billion, respectively.

Altogether, Table S3 shows that the net savings estimates of rescinding the GHG standards decrease from \$490 and \$380 billion to -\$494 and -\$166 billion for discount rates of 3 and 7 percent, respectively.

Table S3: Monetized savings, costs and net benefits of rescinding the GHG standards.

	EPA 2025 DRIA, Scenario 5	Adjustment for AEO 2025 fuel price	Adjustment for AEO 2025 fuel price and our inattention split	Adjustment for AEO 2025 fuel price, our inattention split, and technology costs when attributes can change
	(1)	(2)	(3)	(4)
3 percent discount rate				
Savings	\$1,030	\$1,030	\$1,030	\$597
Costs	\$550	\$642	\$1,092	\$1,092
Net Savings	\$490	\$388	(\$62)	(\$494)
7 percent discount rate				
Savings	\$640	\$640	\$640	\$371
Costs	\$260	\$310	\$538	\$538
Net Savings	\$380	\$330	\$102	(\$166)

Notes: Billions of 2022 dollars. Columns (1) copies Scenario 5 from DRIA Table 3. Column (2) replaces the DRIAs liquid fuel price forecast with those in the AEO 2025 forecast. Column (3) adds our inattention estimate. Column (4) adds an adjustment for technology cost when attributes are allowed to change.

Valuation of environmental externalities

The net benefits of rescinding the standards in Table S4 are further overstated because they ignore externalities from increased carbon emissions and co-pollutants such as local air pollution. Economic theory dictates that such externalities should be priced at their marginal damages. While the DRIA does not quantify emissions impacts, the impact could be large. In the 2024 analysis, the pollution benefits of having the standards amounted to \$1.8 trillion.

B. A Framework for Cost-Benefit Analysis of GHG Vehicle Standards

The following framework is built on the decisions consumers and producers make about fuel economy and the value of environmental gains from reduced pollution. It leads to the core division of the value of fuel savings into the three components discussed in the main text. The framework provides a method for considering costs and benefits and shows which parameters are of most importance.

Prior benefit-cost analyses have revealed that costs and benefits of greenhouse gas standards for fuel economy crucially depend on assumptions regarding consumer preferences and technology costs. By flexibly modeling the consumer's problem, the framework here nests these analyses.

On the vehicle manufacturer side, the framework also importantly allows tradeoffs between fuel economy and other attributes of a vehicle. This component has been missing from much of the regulatory analysis but we show that it is fundamental to understanding valuation and the overall effects of policy. Further, the ability to consider changes in vehicle attributes associated with regulatory compliance has become even more important with the likely use of EVs to meet emissions goals.

B.1 Consumers

Consumers derive benefits from the vehicles they buy. They have a common willingness to pay (WTP) for the vehicle that depends on the present discounted value (PDV) of lifetime fuel costs and a bundle of vehicle attributes other than fuel costs, x . The PDV of lifetime fuel costs is pmg , where p denotes the average price per unit of fuel, m is miles driven, and g is the fuel consumption rate (i.e., gallons per mile equivalent).

Consumers may ignore a share of the PDV of the vehicle's fuel costs, due to inattention. Let $(1 - \alpha)$ indicate the share of the PDV that consumers ignore, and let β indicate WTP for attribute x . The remaining share of lifetime fuel costs that consumers are attentive to is represented by α . The total WTP for the vehicle at the time of purchase is

$$WTP_{purchase} = -\alpha pmg + \beta x.$$

Realized consumer welfare over the life of the vehicle equals the WTP net of the fuel costs the consumer ignores:

$$WTP_{realized} = -\alpha pmg + \beta x - (1 - \alpha)pmg = -pmg + \beta x.$$

Note that we can rewrite the expression for $WTP_{purchase}$ (which we define as component (i) of value in the main text) as:

$$WTP_{purchase} = (1 - \alpha)pmg - pmg + \beta x.$$

Setting pmg equal to -1 (i.e., a regulatory change that saves \$1 in present discounted value of fuel), leads to the relationship described in the main text:

$$\underbrace{WTP_{purchase}}_{\text{component (i)}} + \underbrace{(1 - \alpha)}_{\text{component (ii)}} + \underbrace{(-\beta x)}_{\text{component (iii)}} = 1.$$

B.2 Modeling Technology Improvements without Attribute Trade-Offs

Manufacturers

To examine the effects of the above consumer model in isolation, we first consider a simple manufacturer that sells a single vehicle with fuel consumption rate g and a fixed bundle of other non-fuel cost attributes x . Fuel costs are proportional to the fuel consumption rate. In this case, x is exogenous, and the manufacturer chooses g to maximize profits.

The marginal cost of producing the vehicle is $c = c_0 - \gamma \ln(g)$, where $\gamma > 0$ is the sensitivity of marginal costs to fuel costs.

External costs

Fuel consumption generates external costs in the form of environmental damages. We assume damages are proportional to fuel consumption and equal σmg , where σ denotes the marginal environmental damage per unit of fuel consumption and is abstracted from in our analysis here.

Social welfare, no attribute tradeoff

In this setting, social welfare W is the realized WTP for the vehicle minus production costs and environmental damages:

$$W = -pmg + \beta x - c_0 + \gamma \ln \ln (g) - \sigma mg$$

The welfare-maximizing level of fuel costs is¹

$$g^{\square} = \frac{\gamma}{(p + \sigma)m}$$

Lower technology cost, higher environmental damages, higher fuel prices and higher miles imply a lower optimal fuel consumption rate.

Equilibrium with no regulation, no attribute tradeoff

Suppose sales of each vehicle, q , are exogenous and vehicle price equals consumer WTP at the time of purchase for the vehicle (alternatively, assume perfect competition). With no regulation, the representative manufacturer chooses the fuel consumption rate to maximize profits, which equals sales multiplied by the markup (WTP at the time of purchase minus the marginal cost per vehicle).

$$\{(WTP_{purchase} - c)q = [-\alpha pmg + \beta x - c_0 + \gamma \ln \ln (g)]q\}$$

The first-order condition for the manufacturer problem is

$$g' = \frac{\gamma}{\alpha pm}$$

Comparing this expression to the one for $g^{\square}$ shows that the manufacturer chooses a fuel consumption rate that is higher than the social optimum because of a) undervaluation ($\alpha < 1$) and b) environmental damages.

Fuel consumption rate standard, no attribute tradeoff

¹ Miles traveled could depend on cost per mile as $m = m(pg)$, since miles driven may respond to the price per unit of fuel and the fuel consumption rate. This relationship represents the “rebound effect,” where a lower fuel consumption rate reduces cost per mile, which increases miles driven. Incorporating this into our framework is straightforward and would need a term for the elasticity of miles driven to cost per mile in the welfare expressions (but not in manufacturer decision making).

The manufacturer is subject to a tradable credit standard for fuel consumption rate. To simplify the exposition, we continue to focus on a representative manufacturer.²

The manufacturer is a price-taker in the credit market, and it faces a shadow price per unit of fuel consumption per vehicle of λ , which, by the envelope theorem, represents the marginal cost of tightening the standard. Let $\underline{g}$ indicate the standard. With the standard, the optimization problem is:

$$\left\{ \left[-\alpha p m g + \beta x - c_0 + \gamma \ln \ln (g) + (\underline{g} - g)\lambda \right] q \right\}$$

The manufacturer's first-order condition implies

$$g'' = \frac{\gamma}{\alpha p m + \lambda}$$

Comparing the socially optimal fuel consumption rate $g^{\square}$ and the privately optimal rate with a standard g'' shows that the manufacturer chooses the optimal fuel consumption rate if the stringency of the standard is set so that $\lambda = [(1 - \alpha)p + \sigma]m$. If consumers fully value lifetime fuel costs, then the stringency that incentivizes the manufacturer to choose the socially optimal level of fuel consumption is $\lambda = \sigma m$. The greater the undervaluation, the higher the optimal shadow price and the more stringent the standard.

B.3 Modeling Technology Improvements with Attribute Trade-Offs

Now suppose that reducing the fuel consumption rate simultaneously reduces the other attribute, x . For example, if consumers dislike stop-start ignition, reducing the fuel consumption rate would also reduce the WTP of the vehicle via x and lead to a corresponding reduction in social welfare. As another example, x could represent the disutility from any real or perceived inconvenience of charging an electric vehicle. The EPA refers to this social welfare reduction as a missing, hidden, or opportunity cost associated with attribute tradeoffs. Here, we use the more general terminology of an attribute tradeoff. That is, $\alpha < 1$ is the undervaluation of a pure fuel consumption reduction that does not affect x .

Assume there is a log-linear tradeoff between the fuel consumption rate and the other attribute:

$$-\delta \ln \ln (g_t) + x = 1$$

To distinguish this setting from the setting without an attribute tradeoff, we denote the fuel consumption rate with a t subscript, g_t . The larger is δ , the more that x falls when the manufacturer reduces g_t — that is, the steeper the tradeoff between the two attributes.

After plugging in for x , the social welfare function that accounts for the attribute tradeoff is

² We could allow for heterogeneity across manufacturers in the cost function or consumer preferences, in which case manufacturers would either purchase or sell credits in the market.

$$W_t = -pmg_t + \beta[1 + \delta \ln \ln (g_t)] - c_0 + \gamma \ln \ln (g_t) - \sigma mg_t \quad (1)$$

The optimal fuel consumption rate with the attribute tradeoff is

$$g_t^* = \frac{\gamma + \beta\delta}{(p + \sigma)m}$$

Compared to the optimum with exogenous x , the hidden cost introduces an additional cost of reducing the fuel consumption rate, which is that the manufacturer will also reduce x , harming consumers. The more consumers care about x or the greater the reduction (large δ), the higher the fuel consumption rate.

We note that the functional form allows for a nonlinear tradeoff between g and x . As written, the tradeoff represents a supply-side relationship, but equivalently we could introduce the tradeoff on the consumer side. In this case, the nonlinearity would reflect heterogeneity across consumers regarding their perception of the other attribute. That is, some consumers may perceive larger decreases than others.

Equilibrium with no standard, with attribute tradeoff

Here we consider a case with no standard and with an attribute tradeoff. In this case, the manufacturer solves

$$\{[-\alpha pmg_t + \beta[1 + \delta \ln \ln (g_t)] - c_0 + \gamma \ln \ln (g_t)]q\}.$$

The first-order condition can be expressed as

$$\alpha pm - \frac{\beta\delta}{g_t} = \frac{\gamma}{g_t}.$$

This expression equates the marginal benefit (left-hand side) and the marginal fuel-saving technology cost (right-hand side) of lowering the fuel consumption rate. The marginal benefit includes a lifetime fuel cost term, which is scaled by the parameter α since consumers may undervalue the fuel cost savings when purchasing a vehicle. The marginal benefit is reduced by the magnitude of the marginal consumer welfare loss associated with the attribute tradeoff, which is represented by the second term on the left-hand side. The combined marginal benefit on the left hand side marginal benefit is equivalent to the marginal change in WTP at the time of purchase, which includes consumer valuation of lifetime fuel costs and their valuation of the attribute tradeoff. With no standard, the manufacturer adopts fuel-saving technology up to the point where the marginal value to consumers equals the marginal fuel-saving technology cost.

The first-order condition can be solved for the optimal choice of the fuel consumption rate:

$$g'_t = \frac{\gamma + \beta\delta}{\alpha pm}.$$

Without a standard, the manufacturer chooses a fuel consumption rate that increases with the sensitivity to marginal costs and the attribute tradeoff and decreases with consumer valuation of fuel cost savings.

Fuel consumption rate standard, with attribute tradeoff

With an attribute tradeoff, the optimization problem for a manufacturer facing the standard is

$$\left\{ \left[-\alpha pm g_t + \beta [1 + \delta \ln \ln (g_t)] - c_0 + \gamma \ln \ln (g_t) + (\underline{g} - g_t) \lambda_t \right] q \right\} \quad (2)$$

The manufacturer's first-order condition implies a privately optimal fuel consumption rate:

$$g''_t = \frac{\gamma + \beta\delta}{\alpha pm + \lambda_t}.$$

Several results follow from comparing the socially optimal fuel consumption rate with an attribute tradeoff $g_t^{\square}$ and the privately optimal rate with a standard and an attribute tradeoff g''_t . First, the manufacturer has an incentive to choose the socially optimal fuel consumption rate if the stringency of the standard is set so that $\lambda_t = [(1 - \alpha)p + \sigma]m = \lambda_t^{\square}$. This optimal shadow price is identical to the optimal shadow price without an attribute tradeoff. Note that the equivalence of the shadow prices does not imply that the optimal stringency of the standard ($\underline{g}$) is the same.

Second, when the manufacturer complying with the standard chooses the socially optimal fuel consumption rate (i.e., when the shadow price of the standard equals $\lambda_t = [(1 - \alpha)p + \sigma]m = \lambda_t^{\square}$), the fuel consumption rate is less than the privately optimal fuel consumption rate without a standard: $g''_t < g'_t$.

Third, the existence of attribute tradeoffs (which create missing costs) does not change the optimal policy prescription, which is to set a fuel consumption rate standard that has a stringency (measured by the shadow price λ) that is proportional to the magnitudes of two market failures: undervaluation (as measured by $(1 - \alpha)$) and environmental damages (as measured by σ).

Fourth, compared to the situation without an attribute tradeoff ($\delta = 0$), achieving a given fuel consumption rate requires a higher value of λ and correspondingly higher welfare costs of compliance.

Fifth, if the fuel consumption rate standard reduces the manufacturer's fuel consumption rate by Δg_t , the costs associated with the attribute tradeoff are $\beta[1 + \delta \ln \ln (\Delta g_t)]$. In the main text, we refer to these costs as falling in component (iii) of overall fuel economy valuation.

Summary Points

- The existence of attribute tradeoffs does not change the optimal policy prescription, which is to set a fuel consumption rate standard stringency that is proportional to the magnitudes of two market failures: undervaluation and environmental damages.
- Our framework yields a closed-form solution for the optimal reduction in the fuel consumption rate as a function of key parameters, including the degree of inattention (α), pollution damage costs (σ), the value of attribute tradeoffs (β and δ), fuel-saving technology adoption costs (γ), fuel prices (p), and miles traveled (m).
- Our framework yields a closed-form solution for the shadow price of an optimal standard, which can be compared to “revealed preference” information such as inferred credit prices.

B.4 Modeling a Fuel Consumption Rate Standard, Multiple Technology Options

So far, we have assumed the manufacturer either can adopt fuel-saving technology without an attribute tradeoff or that it can adopt fuel-saving technology with a tradeoff. Suppose instead that the manufacturer has two technology options. The first has a large sensitivity to fuel consumption rate (large γ , denoted by γ_{nt}) and no tradeoff ($\delta = 0$). The second has a low sensitivity to fuel consumption rate (small γ , denoted by γ_t) and positive tradeoff ($\delta > 0$). We denote parameters associated with these technologies using subscripts nt and t (no tradeoff and tradeoff).

If the manufacturer uses the technology without a tradeoff, the optimization problem is the same as that with exogenous x . The optimal fuel consumption rate is:

$$g''_{nt} = \frac{\gamma_{nt}}{\alpha p m + \lambda_{nt}}.$$

If the manufacturer uses the technology with a tradeoff, the optimal fuel consumption rate is:

$$g''_t = \frac{\gamma_t + \beta \delta_t}{\alpha p m + \lambda_t}.$$

Compared to the expression for the fuel consumption rate with no tradeoff, it is ambiguous whether the manufacturer chooses a higher or lower fuel consumption rate with the technology that includes a tradeoff. For the same compliance shadow price ($\lambda_{nt} = \lambda_t$), which technology the manufacturer chooses depends on the sizes of the difference between γ_{nt} and γ_t and $\beta \delta_t$.

When predicting manufacturer compliance behavior, EPA assumes that fuel-saving technologies create pure fuel consumption reductions. That is, the second technology with a tradeoff is not available. If γ_t is sufficiently low relative to the consumer valuation of the attribute tradeoff, the manufacturer would be better off and achieve higher profits by adopting the tradeoff technology. This choice would also increase social welfare relative to the case where the manufacturer does not choose the tradeoff technology. By constraining manufacturer compliance choices, the EPA approach overestimates compliance costs.

Summary Points

- The manufacturer only chooses the tradeoff technology if the savings in engineering costs exceeds the associated cost of the tradeoff.
- By modeling the manufacturer decision as minimizing only engineering costs of fuel-saving technologies without trade-offs, EPA overestimates compliance costs.

B.5 EPA's 2025 Welfare Analysis in the Context of This Framework

We can use our framework to highlight the assumptions EPA makes when it estimates the welfare effects of changing the stringency of the standards (that is, $\underline{g}$).

In the 2025 DRIA, EPA assumes that environmental damages caused by climate change are zero, so that $\sigma = 0$. EPA models the change in social welfare due to a change in the fuel consumption rate as:

$$\Delta W_e = -\alpha p m \Delta g_e + \beta \Delta x(\Delta g_e) + \gamma \ln \ln (\Delta g_e). \quad (3)$$

The term $\beta \Delta x(\Delta g_e)$ represents the change in welfare associated with the change in vehicle attributes due to a change in the fuel consumption rate. EPA assumes that this value is a constant fraction of lifetime fuel cost savings:

$$\beta \Delta x(\Delta g_e) = \mu p m \Delta g_e. \quad (4)$$

Substituting equation (4) into equation (3) yields

$$\Delta W_e = -\alpha p m \Delta g_e + \mu p m \Delta g_e + \gamma \ln \ln (\Delta g_e).$$

Furthermore, EPA assumes that consumers do not exhibit any inattention that contributes to undervaluation, $\alpha = 1$. This simplifies the change in welfare to

$$\Delta W_e = -pm\Delta g_e + \mu pm\Delta g_e + \gamma \ln \ln (\Delta g_e).$$

EPA claims that although $\alpha = 1$ for a pure fuel economy change (that is, no attribute tradeoff), they are only willing to pay for the first 2.5 years of fuel cost savings for real-world fuel economy improvements (which do include attribute tradeoffs). That is, EPA calibrates μ so that the right-hand-side of equation (4) equals the value of post 2.5 years of fuel cost savings, which implies $\mu \approx 0.79$.

In contrast, our framework implies a change in welfare associated with a change in the fuel consumption rate equal to

$$\Delta W_t = -pm\Delta g_t + \beta\delta \ln \ln (\Delta g_t) + \gamma \ln \ln (\Delta g_t) - \sigma m\Delta g_t.$$

Comparing these equations implies that EPA's modeling results in a biased estimate of the change in social welfare if $\sigma \neq 0$ or if consumer willingness to pay for attribute tradeoffs is not equal to post 2.5 years of fuel cost savings, $\beta\delta \ln \ln (\Delta g_t) \neq \mu pm\Delta g_e$.

Decomposing lifetime fuel cost savings

Given our modeling, we can express lifetime fuel cost savings at the time of purchasing a vehicle as the sum of three parameters that sum to 1.

- i) The portion valued by consumers as willingness to pay (WTP).
- ii) The portion not valued because of inattention (or other behavioral reasons). Consumers in fact still get this part of the fuel savings, it's just not in their WTP at the time of purchase.
- iii) The portion offset by observable and unobservable changes in attributes (e.g. acceleration).³

Table S4 compares these three components in our framework with EPA's analysis.

Table S4: Monetized savings, costs and net benefits of rescinding the GHG standards.

³ Although our discussion has considered the case in which reducing the fuel consumption rate reduces the attribute, the attribute changes could be positive. For example, hybrid or all-electric vehicles may have better acceleration than otherwise similar pure gasoline-powered vehicles.

Portion of lifetime fuel cost savings	EPA, 2025 DRIA	EPA, 2024 RIA	Our framework
Valued by consumers as willingness to pay (WTP)	$\alpha - \mu = 0.21$	$\alpha - \mu = 0.21$	$\alpha - \frac{\beta \delta \ln \ln (\Delta g_t)}{pm \Delta g_t}$
Not valued because of inattention	$1 - \alpha = 0$	$1 - \alpha = 0.79$	$1 - \alpha$
Offset by attribute changes	$\mu = 0.79$	$\mu = 0$	$\frac{\beta \delta \ln \ln (\Delta g_t)}{pm \Delta g_t}$

Implementing this framework in benefit-cost analysis

EPA does not explicitly model consumer behavior, and the manufacturer chooses technologies that minimize costs given assumed markup and sales (the 2024 rule had a reduced-form model that predicted market shares, but doesn't model consumer welfare explicitly). EPA (2024) assumes all technologies are pure fuel-saving technology, and EPA (2025) makes an ex-post adjustment to approximate missing costs.

Our framework shows that this will overstate compliance costs. Approximating missing costs as the difference between the PDV of fuel cost savings and consumer valuation of those savings yields biased welfare estimates since the two are not equal. Even if the EPA assumption that $\mu = \frac{\beta \delta \ln \ln (\Delta g_t)}{pm \Delta g_t}$ held historically, there is no reason why this relationship should hold in the future, with new technologies and evolving consumer preferences.

Implementing our framework would require three changes. The first is to take the average consumer valuation of pure fuel cost savings (e.g., savings not associated with a change in other attributes) from the literature. The second change is to model attribute tradeoffs, at least for major technology options. These tradeoffs are technology-specific. For example, tradeoffs for adopting strong hybrids are different from tradeoffs from adopting an all-electric powertrain. The

third change is that manufacturers account for consumer preferences when choosing compliance strategy.

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